

Answers

Exam in Introduction to economics for sport management

Instructions (read carefully)

- The final exam lasts 3 hours.
- You can use calculators.
- Justify all your answers using economic concepts, theories and reasoning.

Good Luck!

Question 1 (20 points)

Suppose there is a country that can produce two goods: X and Y. The country has 200 machines and 100 workers. Each machine can produce 1 units of X or 2 units of Y. Each worker can produce 4 units of X or 4 units of Y.

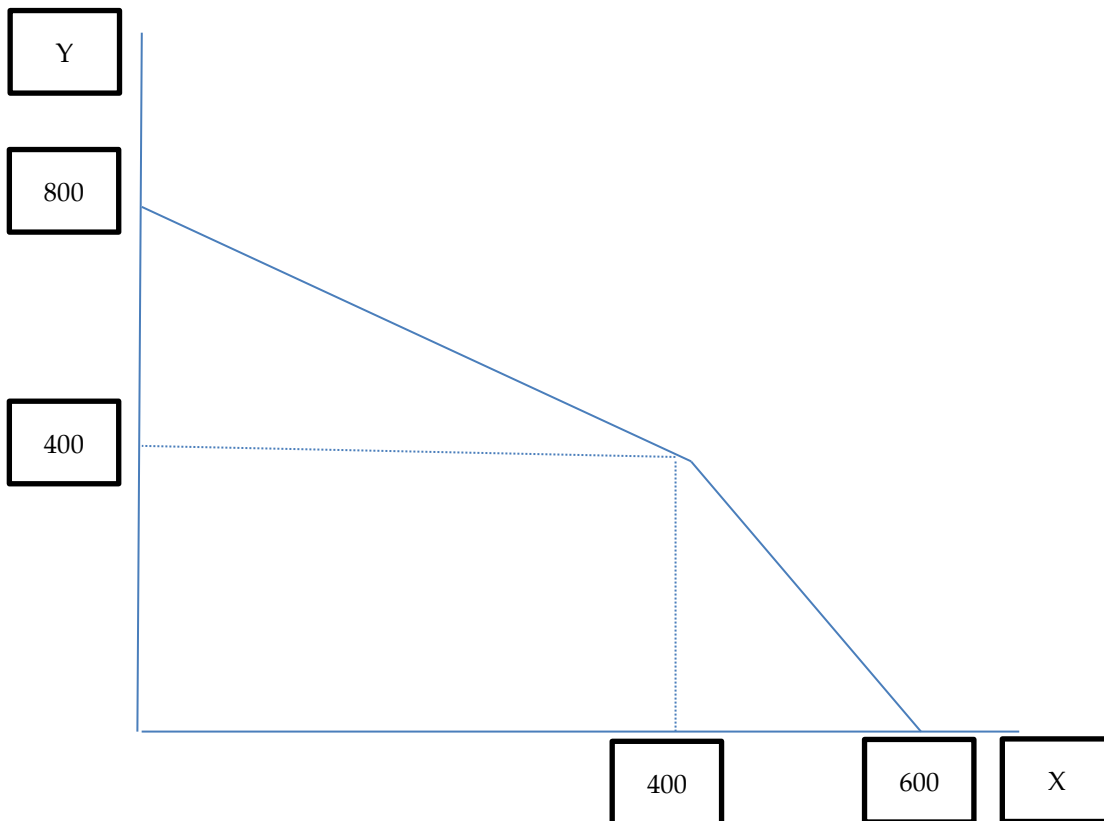
- Please draw a production possibilities frontier of this country and mark all the points (X and Y) where the curve changes its slope as well as the maximal production of X and Y. **(10 points)**
- Suppose, there is a world trade, such that 1 unit of X is worth 3 units of Y. How many units of X and Y should this country produce and why? **(4 points)**
- Suppose that this country needs to consume 500 units of X. In that case how many units of Y will it consume? **(6 points)**

Answers:

	machines	workers
Production X per one unit	1	4
Production Y per one unit	2	4
MC in producing X in terms of Y	2	1
MC in producing Y in terms of X	0.5	1
Max X	$200 \cdot 1 = 200$	$100 \cdot 4 = 400$
Max Y	$200 \cdot 2 = 400$	$100 \cdot 4 = 400$

- See answer a below.

- b. This country has a comparative advantage in producing X, because in the worst case, its cost of producing X is 2Y. However they can sell it to the world for the price of $1X=3Y$. Therefore, this country will only produce 600X and zero Y.
- c. If a country needs to consume 500 units of X, then it will still produce 600 units of X. The country will exchange the remaining 100 units of X for $100 \cdot 3 = 300$ units of Y.



Question 2 (20 points)

There is a market with 8 workers who should be allocated between two fields with the following production function:

Field A

L	TP
0	0
1	5
2	9
3	12
4	14
5	15

Field B

L	TP
0	0
1	3
2	7
3	10
4	12
5	13

- How many workers would work on each field if it is known that the price of one produced unit on field A is 2USD and the price of one produced unit on field B is 1USD? **(6 points)**
- What would be the salary of each worker? **(6 points)**
- What would be the profit of the owners of each field? **(8 points)**

Answer:

Field A

L	TP	mp	Vmp (p*mp)
0	0		
1	5	5	10
2	9	4	8
3	12	3	6
4	14	2	4
5	15	1	2

Field B

L	TP	mp	Vmp (p*mp)
0	0		
1	3	3	3
2	7	4	4
3	10	3	3
4	12	2	2
5	13	1	1

- There will be 4 workers in Field A and 4 workers in field B (or 5 workers on field A and 3 workers on field B).
- The value of the marginal product of the last worker is 2 and this will be the wage as well (VMP=W).
- Profit(A with 4 workers)= $TP \cdot p - W \cdot L = 14 \cdot 2 - 2 \cdot 4 = 20$

$$\text{Profit(A with 5 workers)} = TP \cdot p - W \cdot L = 15 \cdot 2 - 2 \cdot 5 = 20$$

$$\text{Profit(B with 4 workers)} = TP \cdot p - W \cdot L = 12 \cdot 1 - 2 \cdot 4 = 4$$

$$\text{Profit(B with 3 workers)} = TP \cdot p - W \cdot L = 10 \cdot 1 - 2 \cdot 3 = 4$$

Question 3 (20 points)

Suppose the following demand function: $Q^D = 30 - 4p$. In addition suppose the following supply function: $Q^S = 2p$ where Q^D is the quantity demanded of a good, Q^S is the supplied quantity of a good and p is the price of a good in USD.

- What would be the price and the quantity in the equilibrium? (4 points)
- Show on the graph (where appropriate) the cases where $p=0$, $Q=0$, and price and quantity in equilibrium. (8 points)
- What is the producer surplus? (3 points)
- What is consumer surplus? (3 points)
- What is the total social welfare? (2 points)

Answer:

- In equilibrium the supply is equal to demand ($Q^S = Q^D$) therefore, $30 - 4p = 2p$. Solving this equation we get $p=5$ and $Q=10$.

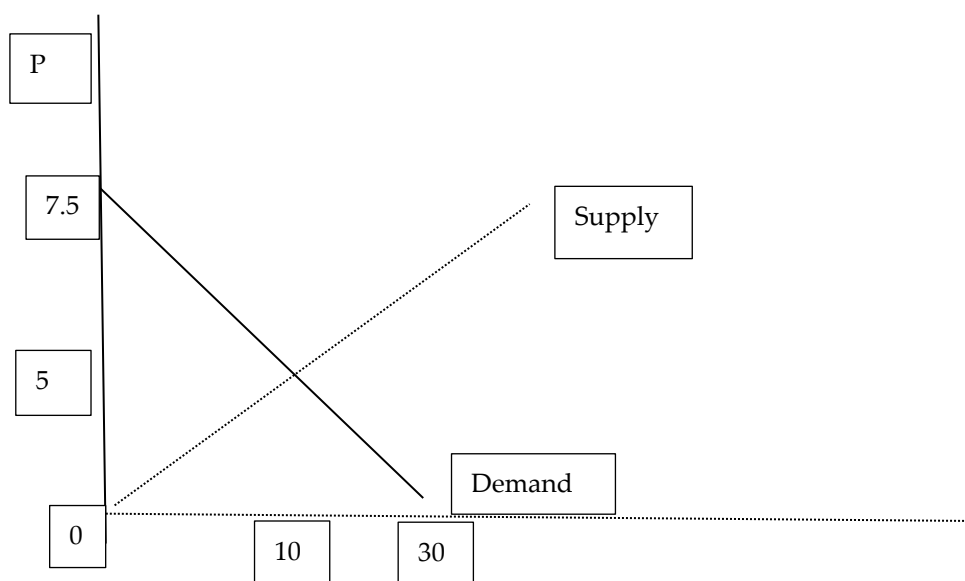
- See the answer below

- c,d,e The total social welfare is the sum of consumer and producer surplus.

The consumer surplus is the area between the demand curve and the price (triangle area) $(7.5 - 5) \cdot (10 - 0) / 2 = 12.5$

The producer surplus is the area between the supply curve and the price (triangle area) $(5 - 0) \cdot (10 - 0) / 2 = 25$

The total social welfare is $25 + 12.5 = 37.5$



Question 4 (20 points)

Assume a firm with the following production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				20			
1	8							
2	15							
3	27							
4	40							

It is known that one unit of labor (L) costs 20 USD.

- Complete the table. **(10 points)**
- What is the minimal price that the firm will produce in the short term? **(2 points)**
- What is the minimal price that the firm will produce in the long term? **(2 points)**
- It is known that the price is 240 USD. What would be the production of the firm? **(2 points)**
- What would be its profit in the short term? **(2 points)**
- What would be its profit in the long term? **(2 points)**

Answer:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0		20		20			
1	8	1/8	20	160	180	160	180	160
2	15	1/7	20	300	320	140	160	150
3	27	1/12	20	540	560	240	186.7	180
4	40	1/13	20	800	820	260	205	200

- min AVC=150. Below that price the firm will not produce in the short term**
- min ATC=160. Below that price the firm will not produce in the long term**
- The rule of production is $mc=p$. Therefore, if $p=240$, the firm will produce 3 units of Q.**
- Profit (short)= $P \cdot Q - VC = 240 \cdot 3 - 540 = 180$**
- Profit (long)= $P \cdot Q - TC = 240 \cdot 3 - 560 = 160$**

Question 5 (5 points)

You are expected to receive a one-time sum of 5000 NOK at the end of the fourth year from now and 3000 NOK at the end of the third year from now. What is the Present Value of this cash flow if the annual interest rate is 8% in the first year, 5% in the second year, 8% in the third year and 4% in the fourth year?

Answer:

$$PV = 5000 / ((1.08)^2 \cdot (1.05) \cdot (1.04)) + 3000 / ((1.08)^2 \cdot (1.05)) = 6375$$

Question 6 (5 points)

You save 1000 NOK at the beginning of each month. What is the Future Value of this cash flow at the end of three months if the monthly interest rate is 1%?

Answer:

$$FV = 1000 \cdot (1.01)^3 + 1000 \cdot (1.01)^2 + 1000 \cdot (1.01) = 3060$$

Question 7 (5 points)

Suppose an inelastic good X. Now suppose that the price of that good has decreased. What will happen to the total revenues of the suppliers?

Answer:

A decrease in the price of a good with inelastic demand will decrease the total expenditure of consumers (total revenue of suppliers).

Question 8 (5 points)

Provide a list of five possible gains from hosting Mega-Events.

Answer from the presentation (6 gains):

- **General infrastructure that are politically difficult to provide without the strict deadline of an opening ceremony (roads, airports, etc.).**
- **An increased number of tourists after the event.**
- **Promotion of a higher sports participation (health).**
- **New stadiums.**
- **Signal to investors that a country is ready to welcome foreign investments.**
- **“Putting a country on the map” (politically and economically).**